

Tests for alcohols

Alcohols are neutral compounds.

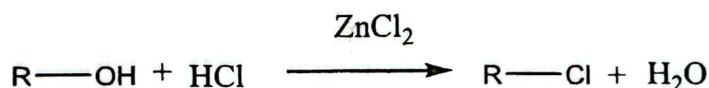
1. Chromic acid test

This test is based on the reduction of chromium (VI), which is orange, to chromium (III), which is green, when an alcohol is oxidized by the reagent. Primary alcohols are oxidized to carboxylic acids and secondary alcohols are oxidized to ketones.

Procedure: Dissolve the unknown liquid (10 drops) or the unknown solid (10 mg) in acetone (1 ml). Add one drop of chromic acid reagent and note the result that occurs within 2 seconds. A positive test for a primary or a secondary alcohol is the appearance of a blue-green color. Tertiary alcohols do not give the test within two seconds, and the solution remains orange.

2. Lucas test

This test depends on the appearance of an alkyl chloride as an insoluble second layer when an alcohol is treated with a mixture of HCl and ZnCl₂ (Lucas reagent)



Primary alcohols do not react at room temperature and therefore the alcohol is seen simply to dissolve. Secondary alcohols react slowly, whereas tertiary, benzylic, and allylic alcohols react instantly.

This test works only for alcohols that are soluble in the reagent. This often means that alcohol with more than six carbon atoms cannot be tested.

Procedure: cool 1 ml of conc. HCl in an ice bath and dissolve 1 g of ZnCl₂ in it, add 2-3 drops of the unknown compound to this solution and shake vigorously. Tertiary benzylic and allylic alcohols give an immediate cloudiness in the solution as the insoluble alkyl halide separates from the aqueous solution. After a short time, the immiscible alkyl halide will form a separate layer. Secondary alcohols produce a cloudiness after 2-5 minutes. Primary alcohols dissolve in the reagent giving a clear solution.

Tests for esters

Note the characteristic pleasant fruity odor.

1. Basic hydrolysis:

Procedure: boil a little of the unknown with 1-2 ml of dilute aq. NaOH for 5-10 minutes. (Add a boiling stone) Stop heating and observe the solution to determine whether the oily ester layer has disappeared or whether the pleasant odor of the ester has disappeared. Acidify with conc. HCl to see whether any precipitate is formed. Precipitate indicates an aromatic acid.

2. Ferric hydroxamate test

On being heated with NH_2OH esters (and lactones) are converted to corresponding hydroxamic acid (N-hydroxyamides) which form strong, colored complexes with Fe^{3+} ions in acidic medium.

Procedure: Dissolve 2-3 drops of the liquid unknown, or about 40 mg of a solid unknown, in a mixture of 0.5 ml of 25% hydroxylamine hydrochloride solution and 1 ml of aq. NaOH. Heat the mixture to boiling for a few minutes. Cool the solution and then acidify with dilute HCl. If the solution becomes cloudy, add 2 ml of 95% ethanol to clarify it. Add a drop of FeCl_3 solution and note whether a color produced. If the color fades, continue to add ferric chloride until the color persists. A positive test should give a deep red or violet color.

